

Healthcare Access, Emergency Assistance & Hospital Transparency Platform

A unified digital platform for finding hospitals, doctors, beds, blood, ambulances and healthcare services —while improving

B.Tech IT Project Presentation

1. Problem Statement

Key healthcare access and transparency challenges addressed by the project

Patients may not receive treatment on time.

Nearby hospitals may have no easy-to-check bed or ICU information.

Ambulance/emergency vehicle availability can be difficult to find.

Blood and other emergency resources may be difficult to locate quickly.

Patients may not know which hospital provides a required facility.

Healthcare charges and unexpected/excessive charges can be difficult to understand.

Users may struggle to report hospital-related problems and track complaints.

Less-educated users and people in poor-connectivity areas need simpler access.

2. Project Objectives

What the proposed platform aims to enable

Fast Access

Help users quickly discover suitable healthcare facilities and emergency resources.

Transparency

Show declared service prices, verification status and update timestamps.

Emergency Support

Connect users with hospitals, beds, blood and emergency vehicles.

Complaint Tracking

Provide evidence-based reporting with a traceable complaint workflow.

Inclusive Access

Support Hindi/English, accessibility, PWA and low-bandwidth usage.

Scalable Integration

Prepare architecture for authorized government healthcare data/API integration.

3. System Users & Roles

Role-based access is a core part of the platform

PATIENT / USER

Search, emergency help, appointments, complaints, feedback, saved facilities and notifications.

DOCTOR

Verified profile, availability, appointments, communication and feedback management.

HOSPITAL ADMIN

Hospital profile, beds, facilities, blood, ambulance, services, pricing and appointments.

GOVERNMENT / ADMIN

Verification, monitoring, complaints, pricing, reports, schemes and audit activity.

SUPER ADMIN

System-level management, security controls and administrative configuration.

4. Core Solution

A single entry point for healthcare discovery, emergency assistance and transparency

Find Hospital

Location + filters for distance, beds, ICU, emergency, blood, specialist and schemes.

Find Doctor

Search by specialization, hospital, location, availability, experience and consultation mode.

Emergency Help

Large, fast-access actions for nearest hospital, ambulance, blood, beds and emergency contacts.

Beds & Blood

Availability status with last-updated time, source and verification status.

Appointments

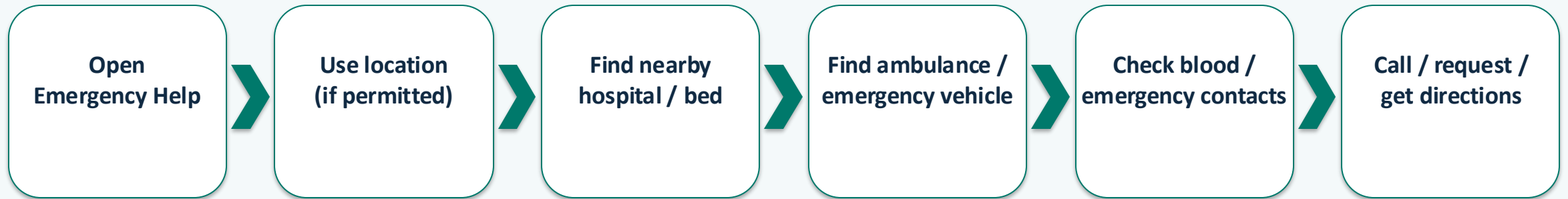
Request/book appointments and communicate during permitted availability.

Report Problem

Submit complaints with bills, images, video, audio or PDF evidence and track status.

5. Emergency Assistance Workflow

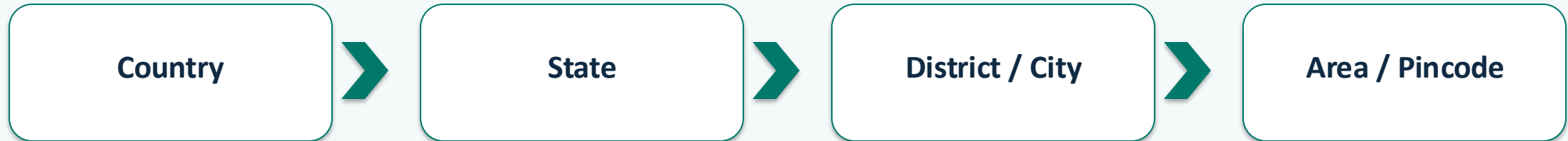
Designed for minimum interaction and fast access to verified information



t: the platform provides information and connects users to healthcare services. It does not independently diagnose diseases or prescribe tre

6. Location-Based Search & Discovery

Hierarchical location + optional device location



After location selection

Hospitals • Clinics • Doctors • Emergency Services • Blood Banks • Ambulances • Beds • Healthcare Facilities

Use My Location

Optional browser/device location permission; location access is never forced.

7. Hospital Transparency & Complaint System

Transparency without automatically accusing or judging a healthcare provider

Price Transparency

Declared prices for consultation, rooms, ICU, diagnostics, blood-related services, medicines/products where applicable and procedures.

Pricing Concern

Users can report unexpected/excessive charges and attach bills or other evidence for authorized investigation.

Complaint Categories

Charges, facility availability, beds, doctor-related concerns, staff behavior, emergency/ambulance/blood issues and incorrect information.

Complaint Lifecycle

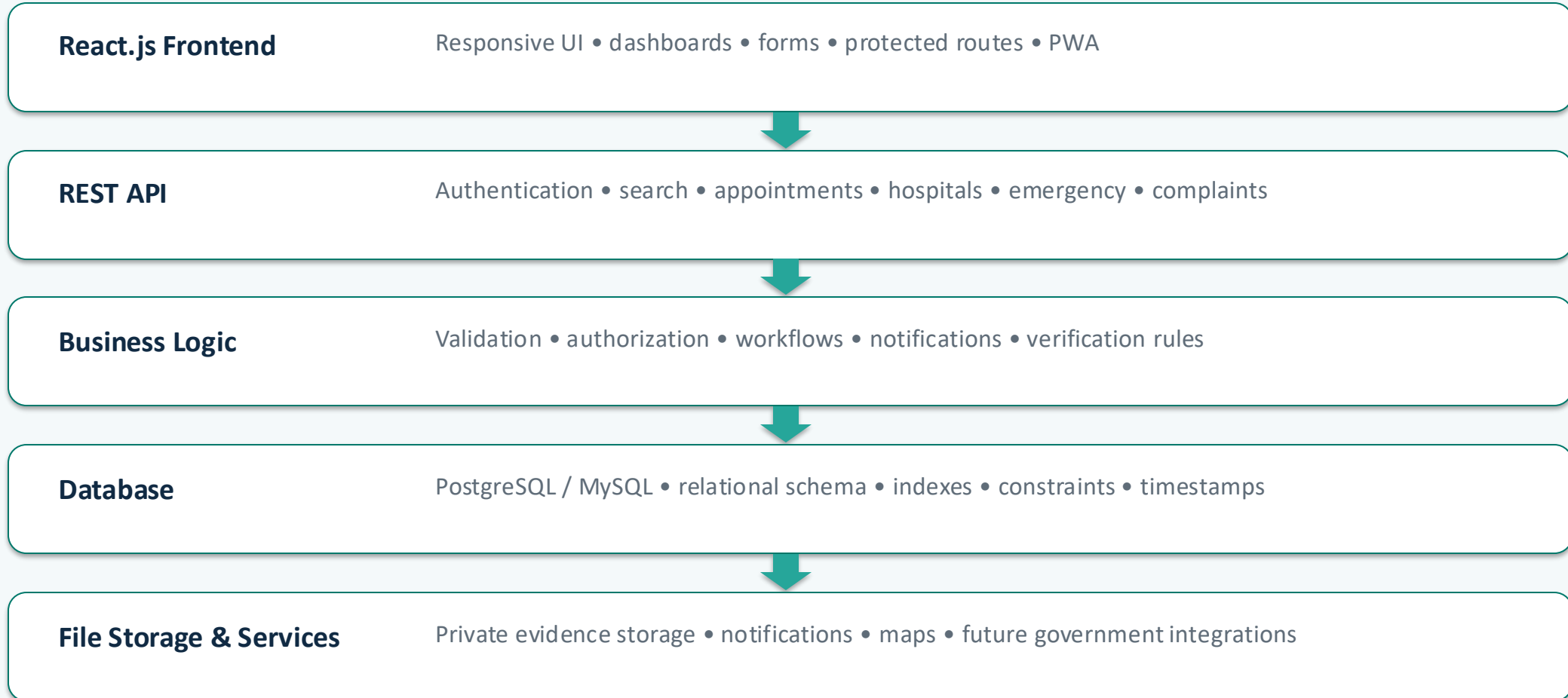
SUBMITTED → UNDER REVIEW → ASSIGNED → INVESTIGATION → RESOLVED → CLOSED

Unique Complaint ID

Example: HC-2026-000123. Users can track status and authorities can maintain investigation history.

8. System Architecture

Production-oriented layered architecture



9. Technology Stack

Suggested implementation stack from the project specification

Frontend

React.js
React Router • API integration • PWA

Backend

Node.js + Express.js
OR Java + Spring Boot

Database

PostgreSQL / MySQL
Relational data model

Security

JWT/access + refresh tokens
Hashing • RBAC • rate limiting

Offline

Service Worker • Cache Storage
IndexedDB • background sync

Integration

Maps • Email/SMS/Push
Future authorized government APIs

10. Database Design

Relational data model supporting the major platform workflows

Identity

users • roles • patients • doctors •
doctor_verifications

Hospitals

hospitals • hospital_admins • departments •
facilities

Resources

beds • bed_availability • blood_banks •
blood_inventory • ambulances

Services

appointments • doctor_availability •
hospital_services • service_prices

Trust & Reporting

reviews • complaints • complaint_evidence •
complaint_status_history

Platform

notifications • locations •
government_schemes • audit_logs •
emergency_contacts

11. Security, Privacy & Access Control

Healthcare data requires strong security and controlled access

HTTPS / encryption in transit and encryption at rest where appropriate.

Secure authentication, password hashing, session management and logout from all devices.

Role-Based Access Control for Patient, Doctor, Hospital Admin, Government Admin and Super Admin.

Input validation, rate limiting, secure headers, SQL injection and XSS protection.

Private evidence storage with MIME/extension validation, file-size limits and malware-scanning architecture.

Audit logs and data minimization; sensitive patient information must not be publicly exposed.

Patient medical records should only be accessible through an authorized healthcare workflow.

12. PWA, Offline & Low-Bandwidth Support

A key differentiator for users with unreliable connectivity

Offline-First

Service Worker + Cache Storage + IndexedDB + background synchronization.

Cached Information

Emergency contacts, saved hospitals/doctors, downloaded area information, profile and pending forms.

Sync Pending

Complaints, appointment requests, feedback, emergency requests and profile changes synchronize when online.

Low Data Mode

Compressed images/JSON, lazy loading, pagination, fewer API calls and text-first emergency screens.

Freshness Warning

Cached bed/ambulance/hospital availability must show its last update time and never be presented as current real-time data.

13. Government / Admin Dashboard

Centralized verification, monitoring, investigation and reporting

Verify doctors and hospitals

Monitor beds, ICU, blood and emergency vehicles

Review and assign complaints

Monitor declared hospital pricing

Manage regions and healthcare facilities

Publish government healthcare schemes and announcements

Generate reports and maintain audit logs

Suspend fraudulent accounts where authorized

14. Government & External Integration Architecture

Designed for future integration without using fake government APIs

GovernmentService

Integration layer

Hospital Data

Hospital information

Doctor Verification

Professional
verification

Emergency Service

Transport /
emergency data

Blood Availability

Authorized inventory
data

Healthcare Schemes

Government scheme
information

Development should use clearly marked DEMO DATA / mock providers. Production should connect only to officially authorized data sources and APIs.

15. Development Roadmap

Incremental implementation plan

1

Authentication + DB + roles

2

Patient + doctor profiles

3

Hospital discovery

4

Beds + blood + facilities

5

Appointments

6

Emergency / ambulance

7

Complaint + evidence

8

Government/admin
dashboard

9

PWA + offline mode

10

Security + testing

11

Deployment

16. Testing Strategy

Functional, integration and security testing

Frontend

Components • forms • validation • routing

Backend

Unit tests • APIs • authentication • authorization

Integration

Registration→Login • hospital search • appointment • complaint •
evidence upload

Offline Sync

Offline action → pending state → online synchronization

Security

Unauthorized access • role escalation • file validation • SQL injection •
XSS • rate limiting

17. Final Product Goal

“Where can I get the healthcare service I need right now?”

✓ Nearby hospitals & verified doctors

✓ Available beds / ICU & blood

✓ Emergency vehicles & contacts

✓ Appointments & healthcare facilities

✓ Transparent service information

✓ Complaint & feedback tracking

✓ Government healthcare schemes

✓ Useful even with poor internet

18. Conclusion

Project summary

Unified Platform

Combines healthcare discovery, emergency assistance, resource availability, appointments and reporting.

Trust & Transparency

Verification, timestamps, declared prices, complaint workflows and audit logs support informed access.

Inclusive Design

Mobile-first, accessible, Hindi/English, PWA, offline-first and low-bandwidth support.

Scalable Foundation

Layered APIs and integration abstractions allow future authorized government and service-provider integrations.

THANK YOU

Healthcare Access • Emergency Assistance • Hospital Transparency